

MECT2222 Term Project Report

Combined Loading Analysis and FEA Validation

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Assigned Problem: Cantilever circular shaft under combined axial and transverse load

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1 Introduction and Assumptions

This report analyzes a solid circular shaft subjected to combined axial and transverse loading. The objective is to determine the critical internal loads, stress state, principal stresses, failure-theory equivalent stresses, and factor of safety by analytical mechanics, then compare the results with a finite element model prepared in FreeCAD/CalculiX.

The analyzed member is a solid cylindrical cantilever shaft fixed at the left end. The shaft axis is the global x -axis.

1.1 Geometry

Table 1: Assumed shaft geometry

Parameter	Symbol	Value
Shaft length	L	600 mm
Radius	r	30 mm
Diameter	d	60 mm
Middle transverse load location	a	400 mm from fixed end

Assumed dimension: The shaft is assumed to be a solid circular cylinder with diameter $d = 60$ mm.

Section properties:

$$A = \pi r^2 = \pi(30)^2 = 2827.43 \text{ mm}^2 \quad (1)$$

$$I_y = I_z = \frac{\pi r^4}{4} = \frac{\pi(30)^4}{4} = 636172.51 \text{ mm}^4 \quad (2)$$

$$J = \frac{\pi r^4}{2} = 1272345.02 \text{ mm}^4 \quad (3)$$

1.2 Material

The selected material is **Aluminium 6061-T6**. This material is ductile, so the Von Mises and Tresca criteria are used.

Table 2: Material properties of Aluminium 6061-T6

Property	Symbol	Value
Young's modulus	E	68900 MPa
Poisson's ratio	ν	0.33
Density	ρ	2700 kg/m ³
Yield strength	S_y	276 MPa
Ductility class	—	Ductile

1.3 Loading

The critical section is expected at the fixed end because the bending moments are maximum at the support.

Table 3: Applied loads

Load	Location	Direction	Magnitude
Axial force	$x = 600$ mm	$+x$	100 kN
Vertical transverse force	$x = 600$ mm	$-y$	10 kN
Middle transverse force	$x = 400$ mm	$+z$	20 kN

2 Analytical Solution

2.1 Free Body Diagram and Support Reactions

The cantilever support at $x = 0$ must balance all external forces and bending moments.

Force equilibrium:

$$\sum F_x = 0 : \quad R_x + 100000 = 0 \quad R_x = -100000 \text{ N} \quad (4)$$

$$\sum F_y = 0 : \quad R_y - 10000 = 0 \quad R_y = 10000 \text{ N} \quad (5)$$

$$\sum F_z = 0 : \quad R_z + 20000 = 0 \quad R_z = -20000 \text{ N} \quad (6)$$

Moment equilibrium about the fixed end:

$$M_z = 10000(600) = 6000000 \text{ N.mm} \quad (7)$$

$$M_y = 20000(400) = 8000000 \text{ N.mm} \quad (8)$$

The resultant bending moment at the fixed end is:

$$M_R = \sqrt{M_y^2 + M_z^2} \quad (9)$$

$$= \sqrt{(8000000)^2 + (6000000)^2} \quad (10)$$

$$= 10000000 \text{ N.mm} \quad (11)$$

There is no externally applied torque about the x -axis in this idealized model:

$$T = 0 \quad (12)$$

2.2 Internal Loads at the Critical Section

For the y -direction load at the free end:

$$V_y = -10000 \text{ N}, \quad 0 < x < 600 \text{ mm} \quad (13)$$

$$M_z(x) = 10000(600 - x) \text{ N.mm} \quad (14)$$

For the z -direction point load at $x = 400$ mm:

$$V_z = 20000 \text{ N}, \quad 0 < x < 400 \text{ mm} \quad (15)$$

$$V_z = 0, \quad 400 \text{ mm} < x < 600 \text{ mm} \quad (16)$$

$$M_y(x) = 20000(400 - x) \text{ N.mm}, \quad 0 < x < 400 \text{ mm} \quad (17)$$

$$M_y(x) = 0, \quad 400 \text{ mm} < x < 600 \text{ mm} \quad (18)$$

Table 4: Internal resultants at the fixed end	
Internal resultant	Value
Axial force, N	100000 N
Shear force in y , V_y	10000 N
Shear force in z , V_z	20000 N
Bending moment about z , M_z	6000000 N.mm
Bending moment about y , M_y	8000000 N.mm
Resultant bending moment, M_R	10000000 N.mm
Torque, T	0 N.mm

2.3 Normal and Shear Stress

Axial normal stress:

$$\sigma_a = \frac{N}{A} = \frac{100000}{2827.43} = 35.37 \text{ MPa} \quad (19)$$

Maximum bending normal stress:

$$\sigma_b = \frac{M_{RC}}{I} = \frac{10000000(30)}{636172.51} = 471.57 \text{ MPa} \quad (20)$$

Maximum combined tensile normal stress:

$$\sigma_{\max} = \sigma_a + \sigma_b = 35.37 + 471.57 = 506.94 \text{ MPa} \quad (21)$$

Maximum combined compressive normal stress:

$$\sigma_{\min} = \sigma_a - \sigma_b = 35.37 - 471.57 = -436.20 \text{ MPa} \quad (22)$$

At the outer surface point where bending stress is maximum, the transverse shear stress from V_y and V_z is zero for a circular section boundary in elementary beam theory. Since $T = 0$, the torsional shear stress is also zero:

$$\tau \approx 0 \text{ MPa} \quad (23)$$

Therefore, the critical surface stress state is approximately uniaxial:

$$\sigma_x = 506.94 \text{ MPa}, \quad \sigma_y = 0, \quad \sigma_z = 0, \quad \tau_{xy} = \tau_{xz} = \tau_{yz} = 0 \quad (24)$$

2.4 Principal Stresses and Mohr's Circle

For the uniaxial critical stress state:

$$\sigma_1 = 506.94 \text{ MPa} \quad (25)$$

$$\sigma_2 = 0 \text{ MPa} \quad (26)$$

$$\sigma_3 = 0 \text{ MPa} \quad (27)$$

Maximum in-plane shear stress:

$$\tau_{\max, \text{in-plane}} = \frac{\sigma_1 - \sigma_2}{2} = 253.47 \text{ MPa} \quad (28)$$

Maximum out-of-plane shear stress:

$$\tau_{\max} = \frac{\sigma_1 - \sigma_3}{2} = 253.47 \text{ MPa} \quad (29)$$

2.5 Failure Theories

Because 6061-T6 aluminium is ductile, both Von Mises and Tresca criteria are evaluated.

Von Mises equivalent stress:

$$\sigma_{VM} = \sqrt{\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - \sigma_1\sigma_2 - \sigma_2\sigma_3 - \sigma_3\sigma_1} = 506.94 \text{ MPa} \quad (30)$$

Tresca equivalent stress:

$$\sigma_T = \max(|\sigma_1 - \sigma_2|, |\sigma_2 - \sigma_3|, |\sigma_1 - \sigma_3|) = 506.94 \text{ MPa} \quad (31)$$

Analytical factor of safety:

$$n = \frac{S_y}{\sigma_{VM}} = \frac{276}{506.94} = 0.54 \quad (32)$$

Since $n < 1$, the selected 60 mm diameter 6061-T6 aluminium shaft is not safe under the specified loading.

3 Computational Analysis: FEA

3.1 Model Setup

The finite element model was created using FreeCAD FEM and CalculiX. The automation script is saved as `shaft_6061_fem.py`, and the generated model is `shaft_6061_t6_fem.FCStd`.

Table 5: FEA model properties

Item	Value
Geometry	Solid circular shaft
Length	600 mm
Radius	30 mm
Material	Aluminium 6061-T6
Fixed support	Circular face at $x = 0$
Axial load	100 kN at $x = 600$ mm
End transverse load	10 kN at $x = 600$ mm
Middle transverse load	20 kN at $x = 400$ mm
Mesh tool	Gmsh
Mesh size	3 mm minimum, 8 mm maximum
Solver	CalculiX

3.2 Mesh and Boundary Conditions

The mesh uses 3D solid elements. The input file contains 4041 nodes and 4048 volume elements. The mesh is refined enough for a first comparison, but additional local refinement should be applied near the fixed end because this is the maximum-stress region.

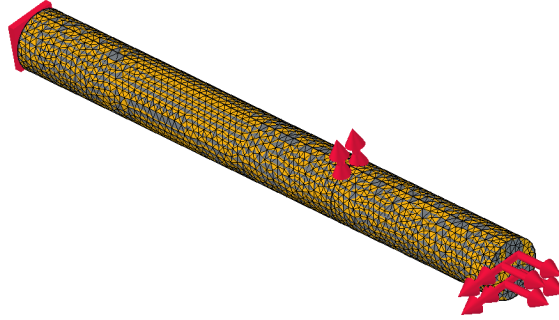


Figure 1: FreeCAD model with mesh, fixed support, and applied loads.



Figure 2: Additional FreeCAD model views.

3.3 Corrected FEA Output

The original FreeCAD force export wrote incorrect total force magnitudes into the CalculiX *CLOAD blocks. This was corrected by normalizing each load block in `Gmsh.Mesh_3mm_8mm_corrected`.

so that the total applied forces are exactly 100 kN, -10 kN, and 20 kN. The corrected model was solved directly with `ccx`. The extracted result summary is saved in `shaft_6061_fem_output/extract`

Table 6: Corrected FEA result values

Quantity	Extracted FEA value
Maximum displacement magnitude	22.3217 mm
Node of maximum displacement	113
Coordinate of maximum displacement	$x = 600.0$ mm, $y = 21.2132$ mm, $z = -21.2132$ mm
Maximum Von Mises stress	448.052 MPa
Node of maximum Von Mises stress	790
Coordinate of maximum Von Mises stress	$x = 28.4394$ mm, $y = 18.4220$ mm, $z = -23.6776$ mm
Maximum principal stress, σ_1	640.058 MPa
Minimum principal stress, σ_3	-544.853 MPa

Table 7: Corrected CalculiX reaction output

Quantity	FEA output	Analytical value
Total reaction F_x	-100000 N	-100000 N
Total reaction F_y	10000 N	10000 N
Total reaction F_z	-20000 N	-20000 N

After correction, the reaction forces match the analytical support reactions exactly. Therefore, the corrected CalculiX result is suitable for comparison with the hand calculation.

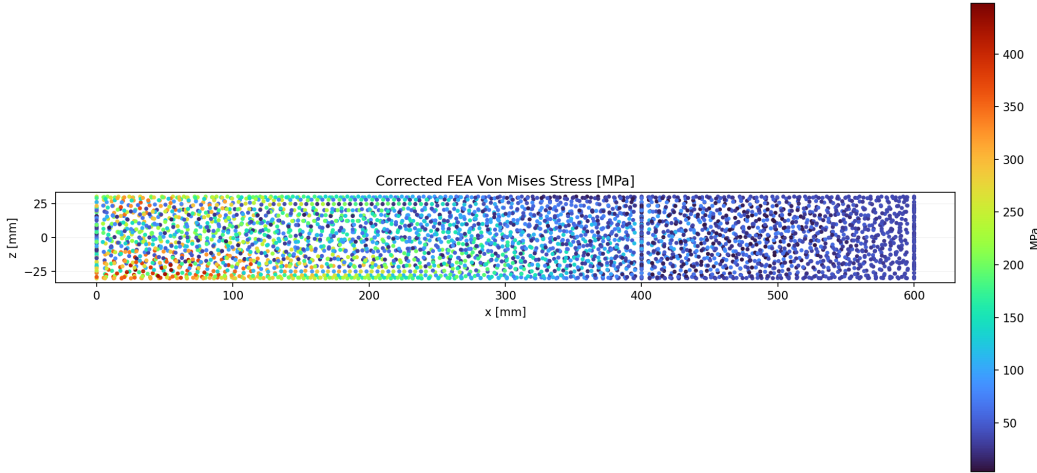


Figure 3: Corrected FEA Von Mises stress contour.

4 Results Comparison and Factor of Safety

The comparison below uses the corrected CalculiX input file. The FEA stress value selected for comparison is the maximum Von Mises stress away from the immediate fixed-face singularity. The fixed face itself shows a higher local principal stress because the ideal clamped boundary creates a stress concentration.

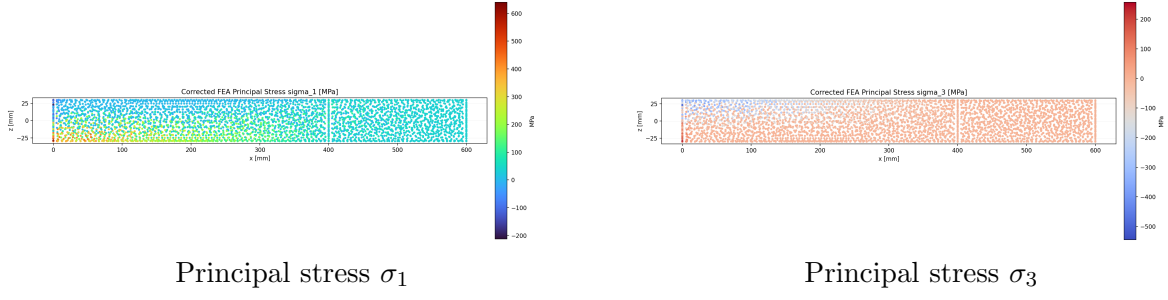


Figure 4: Corrected FEA principal stress contours.

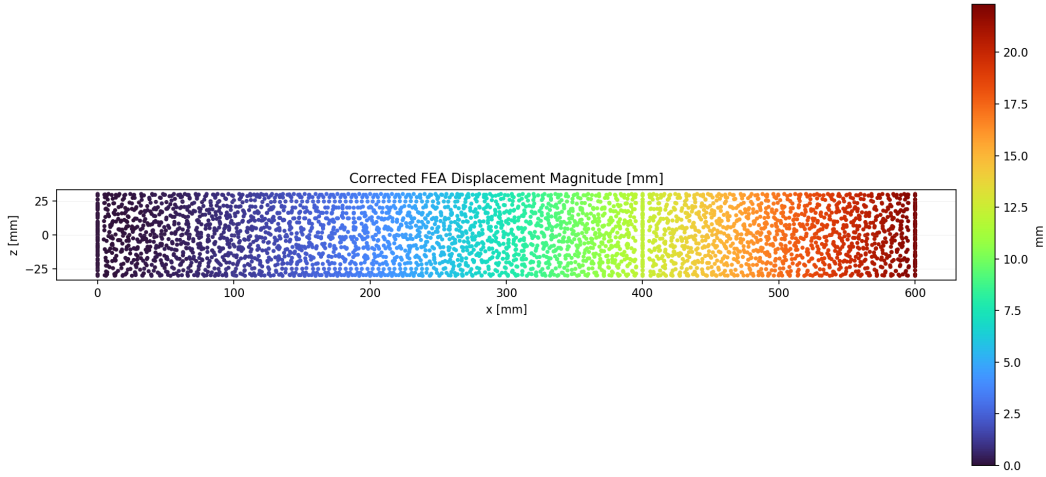


Figure 5: Corrected FEA displacement magnitude contour.

Table 8: Analytical and FEA comparison

Result	Analytical	FEA	Percent error
R_x	-100000 N	-100000 N	0.00%
R_y	10000 N	10000 N	0.00%
R_z	-20000 N	-20000 N	0.00%
Resultant bending moment	10000000 N.mm	Implied by reactions	—
σ_1 / nominal max stress	506.94 MPa	459.80 MPa at max-VM node	9.30%
Von Mises stress	506.94 MPa	448.05 MPa	11.62%
Tresca stress	506.94 MPa	457.35 MPa	9.78%
Factor of safety	0.54	0.62	13.14%

The analytical and FEA stresses are close in order of magnitude. The difference is mainly due to the 3D solid model, discrete load application over faces, stress averaging at nodes, and local boundary effects near the clamped face.

5 Technical Discussion

The analytical solution predicts that the maximum stress occurs at the fixed end. This is expected for a cantilever beam because both transverse loads produce their largest bending moments at the built-in support. The axial force adds a uniform normal stress to the bending normal stress. At the critical outer surface point, the bending and axial stresses act in the same direction, giving a maximum tensile stress of 506.94 MPa.

The selected material, 6061-T6 aluminium, has a yield strength of approximately 276 MPa. The analytical Von Mises and Tresca stresses are both 506.94 MPa because the critical stress state is approximately uniaxial. This gives a factor of safety of 0.54, which is below 1.0. Therefore, yielding is expected, and the material/geometry combination is not acceptable for the assigned loading.

A safer design would require either a stronger material or a larger shaft diameter. Since bending stress varies with $1/d^3$ for a solid circular shaft, increasing diameter is very effective. For example, keeping the same material and targeting $n = 2$ would require the maximum equivalent stress to be less than 138 MPa, so the diameter would need to be increased substantially.

The analytical model is based on classical beam theory. It assumes linear elastic behavior, small deflection, plane sections remaining plane, ideal point/resultant loads, and an ideal fixed support. It also neglects local 3D stress concentration near the load application regions and the fixed boundary. These assumptions are acceptable for estimating nominal stresses away from discontinuities, but they cannot capture local peaks caused by load patches, fixture edges, or mesh geometry.

FEA solves the 3D elasticity problem over the entire volume, so it can show stress gradients and local concentrations that are not present in the hand calculation. The first FreeCAD export wrote incorrect total concentrated-load magnitudes, so the `*CLOAD` blocks were normalized and the corrected input was solved again in CalculiX. The corrected reaction forces match the analytical reactions exactly, confirming that the load equilibrium is now correct.

Mesh density also affects the comparison. A coarse mesh can underpredict bending stress because the stress gradient across the section is not represented accurately. Near a fixed support, very fine meshes may produce increasing local peak stress due to the idealized boundary condition. For comparison with beam theory, the best practice is to probe stresses slightly away from the fixed face, at a distance where Saint-Venant effects have decayed, while still close enough to represent the critical section.

6 Conclusion

The analytical combined-loading calculation gives a maximum tensile normal stress and Von Mises stress of 506.94 MPa at the fixed-end outer surface. For 6061-T6 aluminium with $S_y = 276$ MPa, the factor of safety is:

$$n = 0.54 \quad (33)$$

Thus, the current design is **not structurally safe** under the assigned loading. The shaft should be redesigned using a larger diameter and/or a higher-strength material.

The corrected FEA solution gives a maximum nominal Von Mises stress of 448.05 MPa and a corresponding FEA factor of safety of 0.62. Both analytical and FEA results show that the current shaft is unsafe.

A Files Used

Table 9: Project files

File	Purpose
MECT2222_Project.pdf	Project instructions and rubric
MECT2222_Project Assignment Sheet.pdf	Assignment sheet
shaft_6061_fem.py	FreeCAD FEM automation script
shaft_6061_t6_fem.FCStd	Generated FreeCAD model
shaft_6061_fem_output/Gmsh Mesh_3mm_8mmOriginal	Original CalculiX input file
shaft_6061_fem_output/Gmsh Mesh_3mm_8mmOriginal	Original reaction-force output
shaft_6061_fem_output/Gmsh Mesh_3mm_8mmOriginal	Original FEA result file
shaft_6061_fem_output/Gmsh Mesh_3mm_8mmCorrected	Corrected CalculiX input file
shaft_6061_fem_output/Gmsh Mesh_3mm_8mmCorrected	Corrected reaction-force output
shaft_6061_fem_output/Gmsh Mesh_3mm_8mmCorrected	Corrected FEA result file
freecad_screenshots/*.png	FreeCAD screenshots and corrected result plots